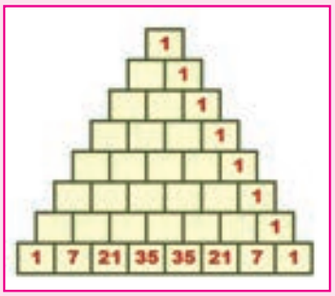
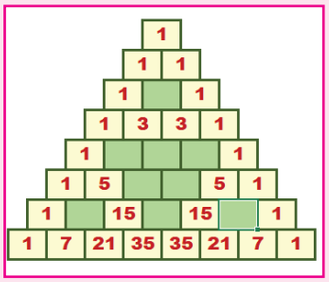
## Try these

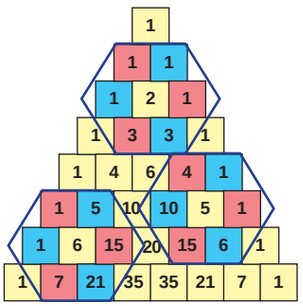
1. Observe the pattern of numbers given in the slanting rows earlier and complete the Pascal's Triangle.

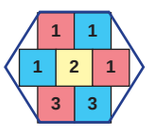


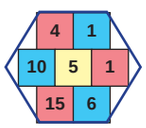

2. Complete the given Pascal's Triangle. Find the common property of the numbers filled by you. Can you relate this pattern with the pattern discussed in situation 2. Discuss.

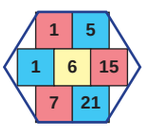



**Example 5.4** Observe the numbers in the hexagonal shape given in the Pascal's Triangle. The product of the alternate three numbers in the hexagon is equal to the product of remaining three numbers. Verify this

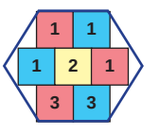
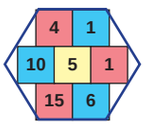
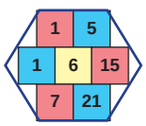



(i) 


(ii) 


(iii) 


***Solution***

| S. No. | Hexagonal Shape | Product of alternate numbers | Product of other three alternate numbers |
|---|---|---|---|
| (i) |  | $1\times1\times3=3$ | $3\times1\times1=3$ |
| | | Both are same | |
| (ii) |  | $1\times6\times10=60$ | $1\times15\times4=60$ |
| | | Both are same | |
| (iii) |  | $1\times5\times21=105$ | $1\times15\times7=105$ |
| | | Both are same | |

## Think

The numbers 1, 3, 6, 10, ... form triangles and are known as triangular numbers. How?



7th_Maths_T2_EM_Chp5_1.indd 96 09-07-2019 13:00:01